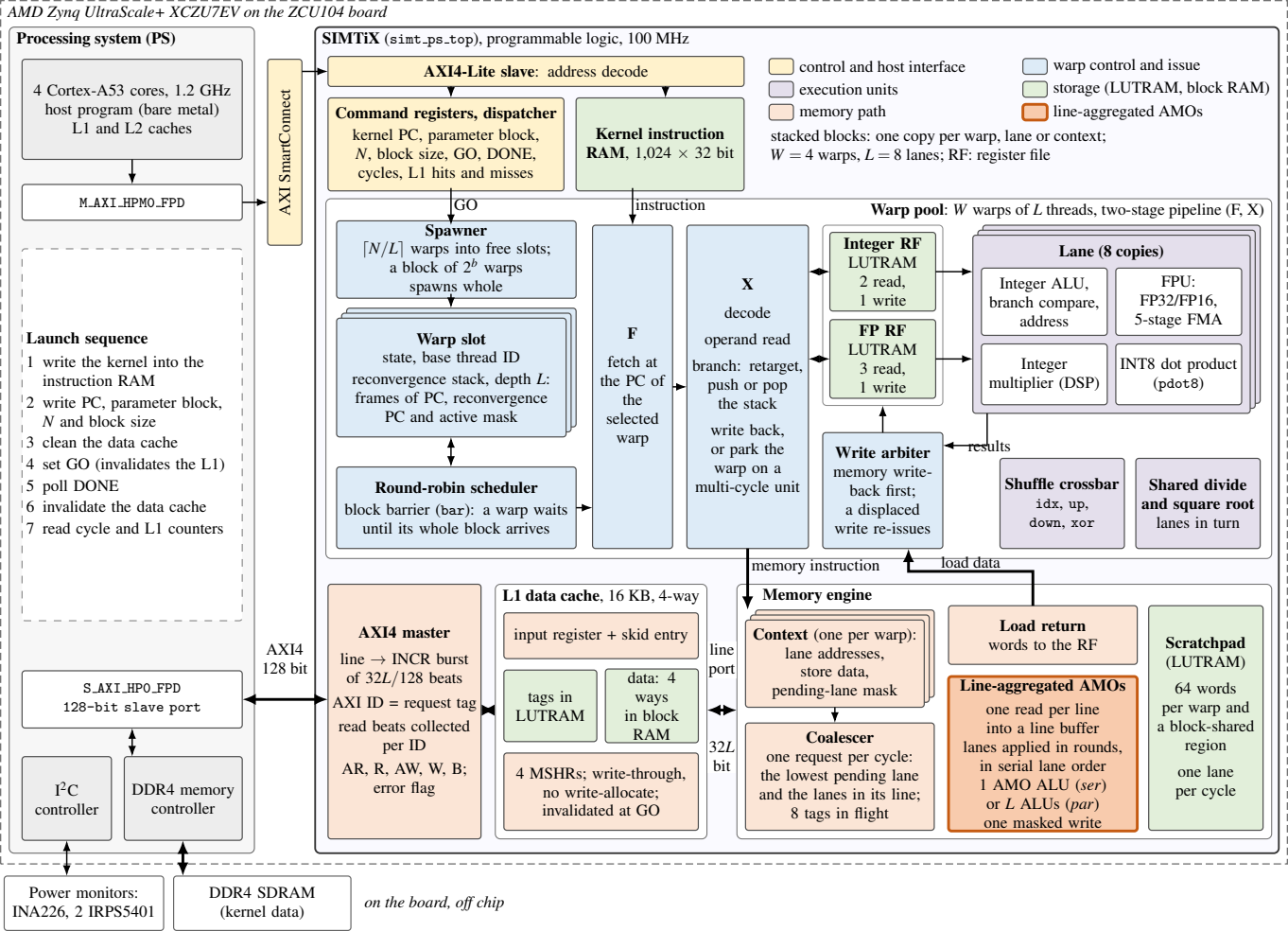

AMD Zynq UltraScale+ XCZU7EV on the ZCU104 board
Processing system (PS)
4 Cortex-A53 cores, 1.2 GHz
host program (bare metal)
L1 and L2 caches
M_AXI_HPM0_FPD
AXI SmartConnect
SIMTiX (simt_ps_top), programmable logic, 100 MHz
AXI4-Lite slave: address decode
Command registers, dispatcher
kernel PC, parameter block,
N, block size, GO, DONE,
cycles, L1 hits and misses
Kernel instruction
RAM, 1,024 × 32 bit
control and host interface
execution units
memory path
warp control and issue
storage (LUTRAM, block RAM)
line-aggregated AMOs
stacked blocks: one copy per warp, lane or context;
W = 4 warps, L = 8 lanes; RF: register file
GO
instruction
Warp pool: W warps of L threads, two-stage pipeline (F, X)
Spawner
⌈N/L⌉ warps into free slots;
a block of 2^b warps
spawns whole
Warp slot
state, base thread ID
reconvergence stack, depth L:
frames of PC, reconvergence
PC and active mask
Round-robin scheduler
block barrier (bar): a warp waits
until its whole block arrives
F
fetch at
the PC of
the
selected
warp
X
decode
operand read
branch: retarget,
push or pop
the stack
write back,
or park the
warp on a
multi-cycle unit
Integer RF
LUTRAM
2 read,
1 write
FP RF
LUTRAM
3 read,
1 write
Lane (8 copies)
Integer ALU,
branch compare,
address
FPU:
FP32/FP16,
5-stage FMA
Integer
multiplier (DSP)
INT8 dot product
(pdot8)
results
Write arbiter
memory write-
back first;
a displaced
write re-issues
Shuffle crossbar
idx, up,
down, xor
Shared divide
and square root
lanes in turn
memory instruction
load data
Launch sequence
1 write the kernel into the instruction RAM
2 write PC, parameter block, N and block size
3 clean the data cache
4 set GO (invalidates the L1)
5 poll DONE
6 invalidate the data cache
7 read cycle and L1 counters
AXI4
128 bit
AXI4 master
line → INCR burst
of 32L/128 beats
AXI ID = request tag
read beats collected
per ID
AR, R, AW, W, B;
error flag
L1 data cache, 16 KB, 4-way
input register + skid entry
tags in
LUTRAM
data: 4
ways
in block
RAM
4 MSHRs; write-through,
no write-allocate;
invalidated at GO
line
port
32L
bit
Memory engine
Context (one per warp):
lane addresses,
store data,
pending-lane mask
Coalescer
one request per cycle:
the lowest pending lane
and the lanes in its line;
8 tags in flight
Load return
words to the RF
Line-aggregated AMOs
one read per line
into a line buffer
lanes applied in rounds,
in serial lane order
1 AMO ALU (ser)
or L ALUs (par)
one masked write
Scratchpad
(LUTRAM)
64 words
per warp and
a block-shared
region
one lane
per cycle
S_AXI_HP0_FPD
128-bit slave port
I²C
controller
DDR4 memory
controller
Power monitors:
INA226, 2 IRPS5401
DDR4 SDRAM
(kernel data)
on the board, off chip